Preeyansh Arora Majoka

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Education

University of British Columbia May 2026

B.A.Sc. Manufacturing Engineering | IMES Award 2021-2026 (Top 10% Faculty of Engineering) | Co-op Program

Relevant Courses: FEM, CFD, Vibration Mechanics, Advanced Manufacturing, Industrial Automation, Software System Design

Technical Skills

Mechanical: Creo, Windchill, SolidWorks, PDM, FEA, FEM using Python, Tolerance Stack-Up, DFM/DFA, GD&T, Ansys, SimScale

Manufacturing & Ops.: Siemens TIA Portal, CODESYS, PLC Programming (LAD/STEP), IFS/SAP ERP, CNC Machining, 3D Printing

Software & Hardware: Python (NumPy/Pandas/Tkinter/Gurobi), MATLAB, Excel (VBA, Power Query), Git, PCB Design (KiCad), Hardware Debugging (Oscilloscope, DMM), Board Bring-up & Validation, LT Spice, Soldering

Professional Experience

Mechatronics Engineer Intern - Product Development Team Jan 2025 – Sep 2025

ARTMS-Telix Pharmaceuticals Limited *Burnaby, BC*

- Designed **20+** tight-tolerance parts and **5+** subassemblies in **SolidWorks**, applying **DFM/DFA** to reduce part count by **20%** and improve assembly efficiency, and applying **GD&T** to ensure manufacturable, accurate assembly
- Developed a 4-layer voltage-controlled nickel electroplating PCB in **KiCad**, optimizing stack-up and grounding to mitigate EMI, regulate deposition rates, and reduce defects by **15%**, safeguarding **$80,000+** in monthly assets
- Minimized radiation and cut install time **75%** by designing a **bayonet backplate** for USC Molecular Imaging Center
- Engineered Gen3 Target Station prototype, reducing volumetric footprint by **70%** to meet Siemens cyclotron constraints
- Validated cooling-finger thermal performance in **SimScale**, maintaining temperature rise within $\pm$ **25 °C** during beam runs

Mechanical Engineer Intern - Vehicle Integration Team May 2024 – Dec 2024

Hexagon Purus Systems Canada *Kelowna, BC*

- Redesigned cab brackets to reduce dimensional variation **25%**, running **WC/RSS** tolerance stack-ups and Monte Carlo simulations in **Excel** and **Creo EZ Tolerance** to isolate dominant contributors (bolt float, weld distortion)
- Prevented structural resonance under **3G** loads and improved tooling access via bracket redesign and **Creo Simulate** FEA
- Consolidated HV phase cable and powertrain CAN harness inventory into **15 standard lengths** across **400+ truck configurations**, utilizing **Creo Cabling** and a **VBA-driven optimization model** to slash part variance
- Automated an **Excel** Production Deviation (PD) tracker for **20 trucks (∼80+ evolving PDs each)**, centralizing build data
- Developed brake and lighting requirements for the Tern RC-8 truck by applying **FMVSS/CMVSS** motor vehicle standards

Design Team & Projects

President & Chassis Lead Jan 2025 – Present

OKMotorrad MotoStudent Team *Kelowna, BC*

- Founded UBCO's electric motorcycle team, recruiting **20+** members and establishing a multi-disciplinary structure
- Built a functional electric dirt bike prototype in **4 months** in under **$200** by repurposing scrap materials and fabricating custom brackets to drive an **1800W** BLDC motor, successfully validating feasibility to fast-track faculty accreditation
- Led frame topology research and prototype iterations in **SolidWorks Weldments**, simulating structural loads in **SolidWorks Simulate** and **Ansys Mechanical**, improving stiffness-to-weight ratio by ∼**15%**

Automation Lead Jan 2026 – Apr 2026

PLC Control & MES Integration for Festo Cellphone Manufacturing Equipment *Kelowna, BC*

- Achieved **100%** end-to-end production traceability across a simulated mobile phone assembly line by building an **ISA-95** compliant **PyQt5-based MES**, integrating a **SQLite** database to manage real-time order execution and audit trails
- Reduced conveyor motor runtimes by **40%** and synchronized hardware with **<100ms** latency by optimizing conveyor logic in **Siemens TIA Portal** and integrating an **asynchronous OPC UA** client enabling **10Hz** constant polling rate
- Slashed HMI memory usage by **87%**, consolidating **40** static screens into **5** dynamic screens using Job Mailbox area pointer

Project Proposer & Manufacturing Lead Sep 2025 – Apr 2026

Target Capsule Feedback Capstone *Kelowna, BC*

- Defined ARTMS-Telix capstone requirements, led client meetings and used **MS Project** to align deliverables and timelines
- Enabled synchronized **1–5 kHz** multi-sensor tracking via a **SolidWorks**-designed T&V bench simulating cyclotron transfer
- Designed and prototyped a 3D-printed sensor-mounted coupler using **SolidWorks** for a **Ø1.5"** pneumatic hose, ensuring airtight splicing and enabling reliable capsule over **50'-100'** of buried hose length